Website Design Specialist ACS (NWE.0F)

COURSE OUTLINE

COURSE TITLE	Advanced Web Programming	**BLOCK**	3
		SECTION	93106
COURSE NUMBER	582-31B-VA	**SEMESTER**	Winter 2026

HOURS	75	**PREREQUISITES**	Dynamic Web Programming (582-21B-VA)
PONDERATION	2-3-3		
CREDITS	2.66 credits	**CO-REQUISITES**	
TARGET COMPETENCY		EHPA: Program a website EHP2: Develop the back end of a website	

MODALITY DETAILS	**DATES**	**SCHEDULE**	
Online - Synchronous	4/22/2026 – 7/23/2026	Tuesdays & Wednesdays	9:00 – 12:00

COURSE DESCRIPTION
In this course, students are introduced to object-oriented programming. They learn how to build robust web applications, how to handle errors, and continue working with database management systems.
COURSE ROLE IN THE PROGRAM
The course **Advanced Web Programming (582-31B)** is part of the Programming competency area and focuses primarily on server-side programming. In this course, students continue developing their skills in the Python programming language, object-oriented programming principles, and different types of web development architectures. The course serves as a transition from the procedural programming studied in **Dynamic Web Programming** to object-oriented programming.

TRANSVERSAL LEARNING
Autonomy in coding and in researching documentation and solutions Rigor Analytical thinking Logical thinking Creativity

TEACHER'S INFORMATION
Kamyar Karimi **Office Hours:** By appointment through MIO
TEACHING METHODS
- Code along sessions - Labs and assignments reflecting on the lecture. - Individual and group projects with qualitative assessments for further evaluation.

REQUIRED TEXTBOOK / ADDITIONAL EXPENSES (if applicable)
No textbook is required; we will be using online official documentation as we progress in the course.

ASSESSMENT AND EVALUATION	
Assessment activity (tentative)	**Percentage (%)**
Assignments	**11%**
Exams	**3 x 13%**
Project	**20%**
LIA	**30%**
Total	**100%**

LEARNING INTEGRATION ASSESSMENT	
Description	Each course includes a Learning Integration Assessment (LIA) that attests to students' achievement of the course-level learning outcome that is the main knowledge, skills, and attitudes to be developed. **Competencies assessed** EHPA: Program a website EHP2: Develop the back end of a website
Percentage	30%

Evaluation criteria	- **Program design**
	  ○ requirements are met
	  ○ program flow is decomposed into manageable, logical pieces

	○ data structures are appropriate ○ common code is unified, not duplicated ○ appropriate algorithms are used, and coded cleanly ○ APIs are used when appropriate ○ custom element behaves consistently with built-in elements ○ no global variables - **Readability** ○ constants are used instead of hard-coded values ○ complex or meaningful expressions are named ○ naming is consistent and descriptive ○ inline comments are used to explain reasoning when necessary ○ documentation is correct ○ code is well formatted - **Version control** ○ commits contain related changes ○ messages are consistent and informative

COURSE CONTENTS
Class Schedule (subject to change)

Week #	***Date(s)***	***Learning Activities***	***Assessments***
1	21-Apr	Introduction; course overview; dynamic web applications; client-server review; Python environment setup	
1	22-Apr	Classes: defining classes, constructors, instances, object state	
2	28-Apr	Methods: instance methods, self, object behaviour	
2	29-Apr	Class Attributes vs instance attributes	
3	5-May	Static methods and class methods; utility behavior and factory patterns	
3	6-May	**Lab**	

4	12-May	Interfaces / abstraction in Python: protocols, abstract base classes, design contracts	
4	13-May	Visibility and encapsulation conventions in Python; public, protected, private naming	**Exam 1**
5	19-May	Properties: getters, setters, managed attributes	*Asynchronous*
5	20-May	Invariants: maintaining valid object state; defensive programming	*Asynchronous*
6	26-May	Enums and constants in application design	*Asynchronous*
6	27-May	**Lab**	*Asynchronous*
7	2-Jun	Inheritance and polymorphism	
7	3-Jun	Exceptions: raising, handling, custom exceptions, application reliability	**Exam 2**
8	9-Jun	Data classes: concise domain models and value objects	
8	10-Jun	Modules and package organization: structuring larger Python applications	
9	16-Jun	**Lab**	
9	17-Jun	Introduction to Flask: routing, request-response cycle, application structure	
10	7-Jul	Blueprints: modular Flask applications	
10	8-Jul	Database: connecting Flask to a DBMS, ORM basics, CRUD operations	**Exam 3**
11	14-Jul	Models: database models, relationships, domain logic	
11	15-Jul	Validations: server-side validation, form validation, data integrity	
12	21-Jul	View decorators: authentication, authorization, reusable route logic, error handlers	
12	22-Jul	Final Interviews	*LIA*

Policy	Full policy
College-level bylaws and policies:	https://www.vaniercollege.qc.ca/bylaws-policies
Continuing Education bylaws and policies:	https://www.vaniercollege.qc.ca/conted/conted-policies
Assignments	All assignments must be submitted by the due date posted on the course platform. Late submissions will incur a penalty of 5% per day unless prior arrangements have been made with the instructor. Exceptional Circumstances: If you encounter unforeseen circumstances (illness, emergencies), you must notify the instructor before the deadline to request an extension. Grading of Late Work: Assignments submitted more than 5 days late without an approved extension will receive a grade of zero.
Class & Learning environment	Class Start Time: Students are expected to log into the online session on time. Participation: Students are expected to actively participate with microphones and cameras on when possible, unless otherwise discussed with the instructor. Respect for Learning Environment: Chat messages, microphone use, and screen sharing must remain respectful and related to the course content. Disruptive behavior will be addressed immediately and may impact participation grades.

Familiarity with and Adherence to Vanier Course Level Policies

It is the student's responsibility to be familiar with and adhere to the Vanier College Academic Policies. The complete policies can be found on the Vanier College website, under **Policies**. Students should pay particular attention to the **Institutional Policy on the Evaluation of Student Achievement (IPESA)**. The IPESA aims to clarify the structures and means that ensure the fair and equitable evaluation of student achievement, including the conditions for learning, assessment, and certification.

Section 11.4 of IPESA provides details on **Academic Integrity**, which is an essential component of any educational institution and all students are expected to uphold the highest standards of academic integrity during their college studies. Failure to do so may result in serious consequences – please see section 11.4.2 on **Academic Misconduct**. There are many **resources** available to support academic integrity in your courses. If you have a problem that you have been unable to resolve by talking with your teacher, please reach out to the **Student Advocate** or to a Faculty Dean's office (A-286, B-208, N-301). For Continuing Education courses, students can visit E-115.

Academic complaints may be defined as grievances related to any matter that is covered in this policy, and which may have an impact on student achievement. It is the College's responsibility to offer a fair and equitable hearing of student complaints. To do so, the College provides students who perceive that there have been mistakes, errors, or breaches of procedure involving any aspect of this policy with clear complaint and recourse processes.

Complaint and recourse processes may be informal or formal. Students are always encouraged to approach their teacher before seeking help from any outside person or body, but other courses of action may be followed depending on the situation.

More details on Academic Complaints and Recourse Mechanisms can be found in the IPESA, chapter 19.

In the case of Academic Complaints, after having discussed with your teacher, if you are still dissatisfied, you may contact the following, in the specified order:

Pedagogical Counsellor ACS: Ye Tian, E118, tiany@vaniercollege.qc.ca, 514 744 7500 X 8377

Director, Continuing Education, Gregory De Luca E123A, delucag@vaniercollege.qc.ca, 514 744 7500 X 7542

Student Advocate, Student Services: Chelsea McVetty, C202C, mcvettyc@vaniercollege.qc.ca, 514 744 7500 X 7635

COURSE COMPETENCIES

Competency EHPA

OBJECTIVE	STANDARD
Statement of the Competency	**Achievement Context**

Program a website.	- Individually or in collaboration with a work team
	- In connection with a website production project
	- Based on:
	  - production specifications, scenarios or storyboards
	  - website mock-ups
	  - digital media
	- Using:
	  - a computer with the appropriate software
	  - documentation

Elements of the Competency	**Performance Criteria**
1 Analyze the request.	1.1 Proper understanding of the request
	1.2 Accurate identification of request parameters
	1.3 Correct algorithm development
	1.4 Correct choice of the design pattern to be used
	1.5 Proper identification of the necessary data
	1.6 Coherent choice of libraries or web frameworks

2	Program the elements of the website.	2.1	Appropriate choice of instructions
		2.2	Efficient source code splitting
		2.3	Logical organization of instructions
		2.4	Adherence to language syntax
		2.5	Adherence of the code to the algorithm
		2.6	Proper use of the application programming interface (API)
3	Resolve programming errors.	3.1	Effective use of debugging tools
		3.2	Complete detection of errors
		3.3	Effective debugging
		3.4	Relevance of bug fixes

4	Check the quality of the programming.	4.1	Proper implementation of validation tests
		4.2	Complete detection of operating errors
		4.3	Proper correction of operating errors
		4.4	Adequate readability of the source code
		4.5	Strict adherence to coding rules
		4.6	Proper functioning of the programming
5	Document programming.	5.1	Clear notation of comments in the source code
		5.2	Proper use of documentation conventions
6	Manage source code versions.	6.1	Proper configuration of the version management system for the project
		6.2	Methodical submission of modified source code
		6.3	Sound management of branches and conflicts

Competency EHP2

OBJECTIVE	STANDARD
Statement of the Competency	**Achievement Context**

Develop the back end of a website.	- Individually or in collaboration with a work team
	- In connection with a website production project
	- Based on:
	  - production specifications, scenarios or storyboards
	  - website mock-ups
	  - digital media
	- Using:
	  - a computer with the appropriate software
	  - documentation

Elements of the Competency		**Performance Criteria**	
1	Process requests from a web browser.	1.1	Proper identification of the type and parameters of the HTTP request
		1.2	Proper use of HTTP request processing functions

2	Interact with an online database.	2.1	Proper use of the database connection functions
		2.2	Proper use of database query functions
		2.3	Proper formulation of queries
		2.4	Proper management of query errors
		2.5	Proper use of data processing functions
		2.6	Rigorous data processing
		2.7	Rigorous implementation of data security methods
3	Respond to web browser requests.	3.1	Proper preparation of the response to an HTTP request
		3.2	Proper use of functionality and integration methods in web interfaces
		3.3	Accurate integration of dynamic elements into the web interface